StoreTMA Test

Detailed Setup Guide (A to Z)

- Local setup (Windows/WSL + Node + env)
- DB + Stripe configuration
- Domain + server deployment (Vercel)
- Visual demo flow + screenshots
- Troubleshooting

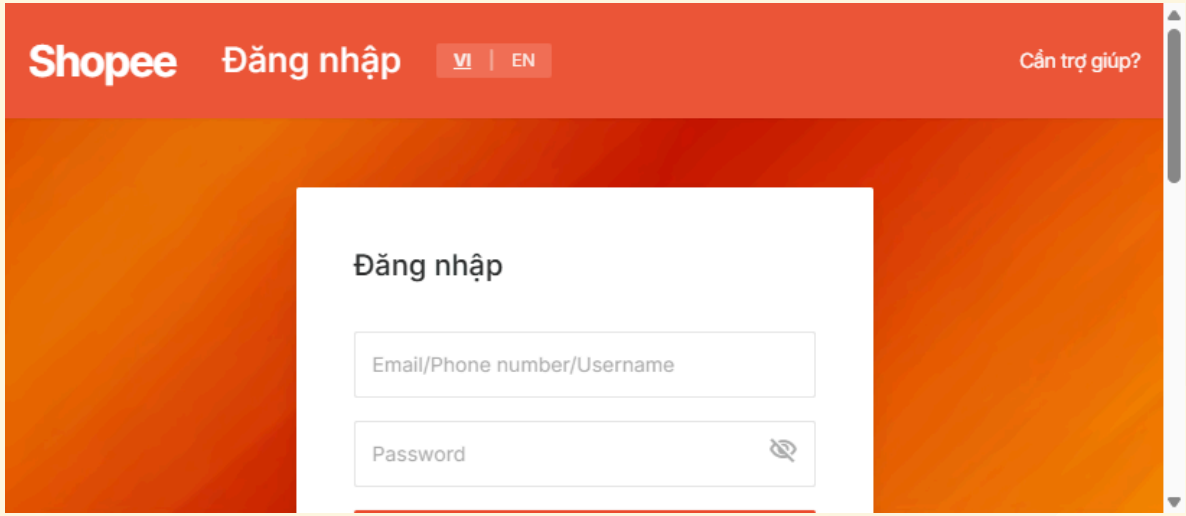
1) Architecture at a Glance

This diagram matches the current repo structure and runtime boundaries.

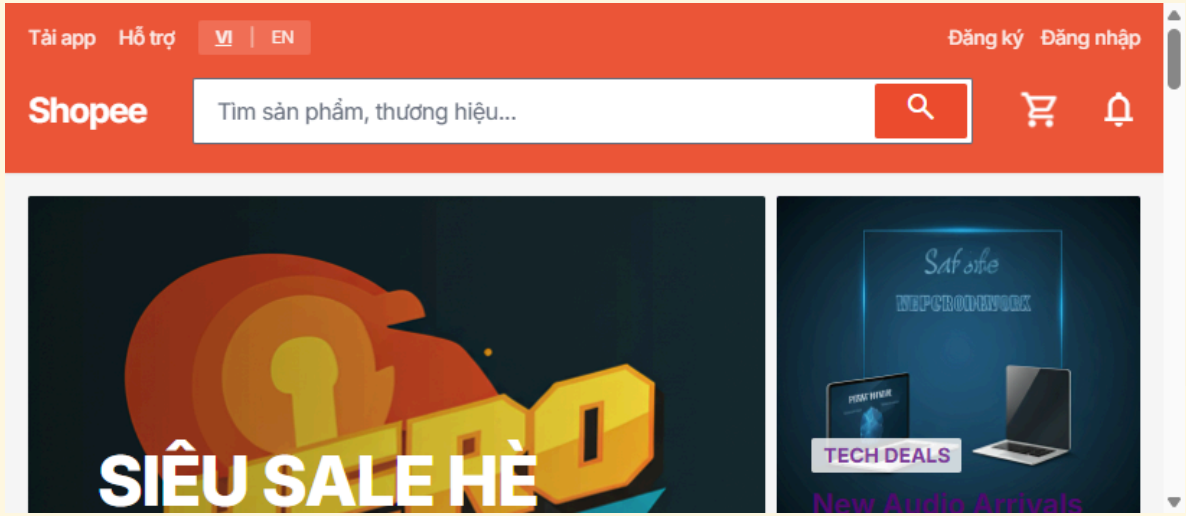
2) Demo Flow (Click-by-Click)

Goal: Home → Add to cart → Checkout → Create Stripe session → Redirect.

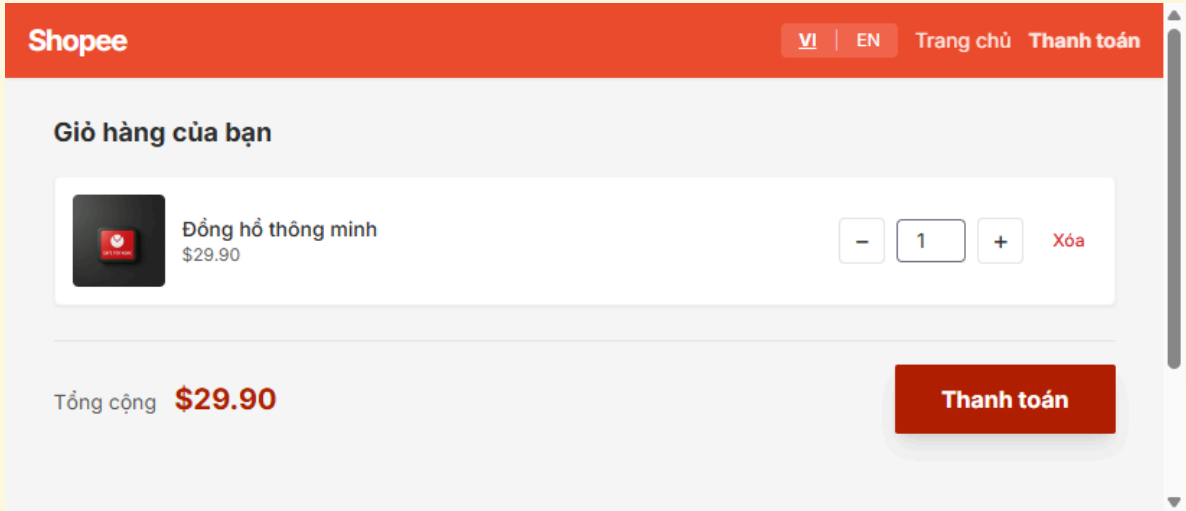
3) Web UI — Login Page



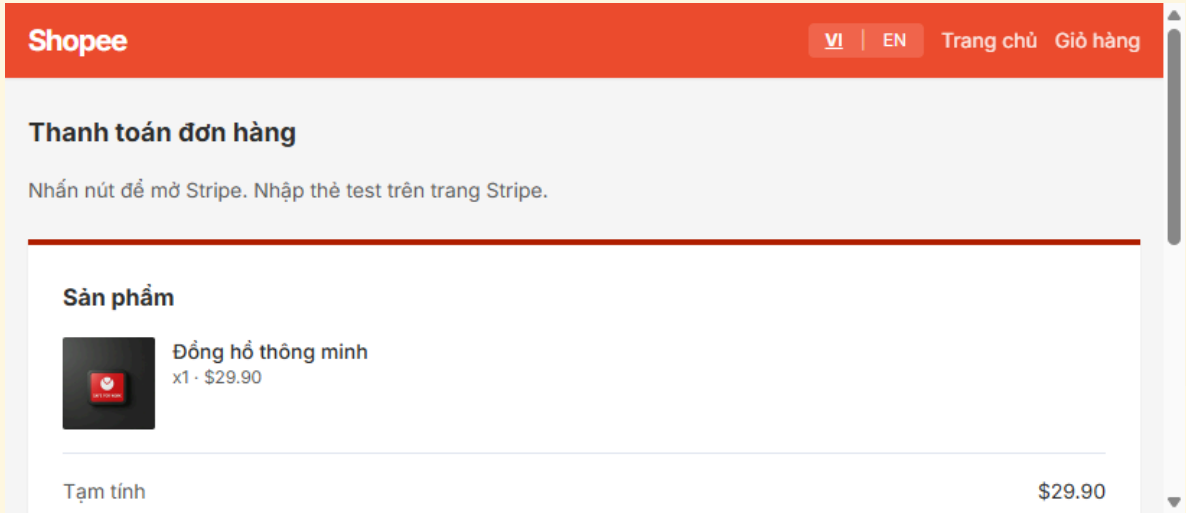
4) Web UI — Home Page



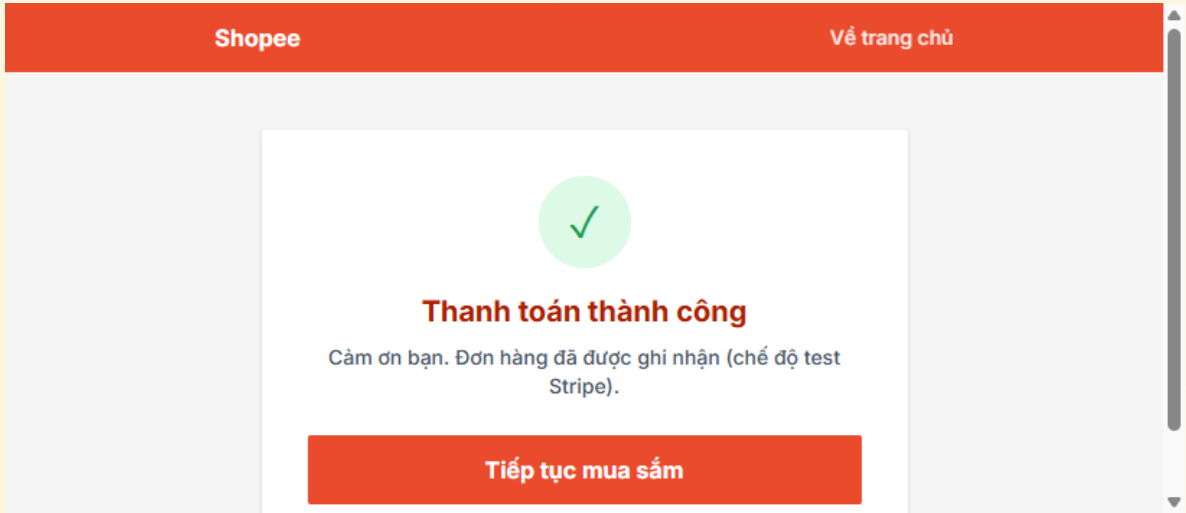
5) Web UI — Cart Page



6) Web UI — Checkout Page



7) Web UI — Success Page



8) Prerequisites (Local)

- Git installed
- Node.js + npm installed
  - Recommended runtime: Node 18/20
  - Supported in this repo: Node 16 (tests include crypto polyfill)
- Stripe account + test secret key (`sk_test_...`)
- Postgres database (Neon recommended)

9) Clone and Install

```
git clone <your-repo-url>
cd storetma_test
npm install
```

Then create `.env.local`:

```
DATABASE_URL="postgresql://USER:PASSWORD@HOST:PORT/DB?sslmode=require"
STRIPE_SECRET_KEY="sk_test_..."
```

10) Database Setup (Products Table)

```sql
CREATE TABLE IF NOT EXISTS products (
  id text PRIMARY KEY,
  name_vi text NOT NULL,
  name_en text NOT NULL,
  price_cents integer NOT NULL CHECK (price_cents >= 0),
  image text NOT NULL,
  section text NOT NULL CHECK (section IN ('flash', 'daily')),
  category text NOT NULL,
  badge text,
  created_at timestamptz NOT NULL DEFAULT now()
);
```

```sql
CREATE INDEX IF NOT EXISTS idx_products_section_created_at
  ON products (section, created_at DESC);
```

11) Run Local Server

```
npm run dev
```

Open:

- `http://localhost:3000/` (rewrites to `login.html`)
- `http://localhost:3000/home`
- `http://localhost:3000/cart`
- `http://localhost:3000/checkout`

API check:

```
curl "http://localhost:3000/api/products?section=daily&limit=5&offset=0"
```

12) Unit Test Commands

```
npm test
```

```
npm run test:watch
```

Notes:

- Tests live in `unittest/`
- Node 16 startup compatibility is handled by:
  - `scripts/crypto-polyfill.cjs`
  - preload in `package.json` test scripts

13) Setup + Domain + Server (Very Detailed)

A) Local environment

1. `npm install`
2. Create `.env.local`
3. `npm run dev`
4. Smoke test pages: `/home` → add to cart → `/checkout`

B) Vercel deploy (server)

1. Import repo into Vercel
2. Set env vars:
   - `DATABASE_URL` (or `POSTGRES_URL`)
   - `STRIPE_SECRET_KEY`
3. Deploy (Preview → Production)

C) Custom domain (DNS)

14) Troubleshooting Matrix

Symptom	Root Cause	Fix
`/api/products` returns 500	DB env missing	Set `DATABASE_URL`, restart app
Checkout 500	Stripe key missing or invalid	Set `STRIPE_SECRET_KEY` with `sk_`
`crypto.getRandomValues` error in tests	Node 16 + Vite startup issue	Keep polyfill preload in test script
Home shows fallback products only	DB/API unreachable	Check DB connectivity and logs

15) Auto-Generate PPTX/PDF

From project root:

```
npm run slides:build
```

Artifacts:

- `docs/slides/dist/setup-detailed.pptx`
- `docs/slides/dist/setup-detailed.pdf`

Slide source: `docs/slides/setup-detailed.marp.md`